

Investigation on the dynamics of Z-pinch in discharge-produced plasma sources via two-dimensional MHD simulations

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ABSTRACT

The plasma Z-pinch plays a pivotal role in generating bright optical radiation with short wavelengths in Discharge-produced Plasma (DPP) sources, such as capillary discharge-produced and inductively driven electrodeless light sources. In this study, we investigate the Z-pinch dynamics of DPP-based electrodeless light sources using two-dimensional magnetohydrodynamic simulations via the USim code. Meanwhile, the influences of plasma density disturbances and the parametric effects of pulsed current on the Z-pinch process are examined and discussed. The results reveal that in the absence of density disturbances, the interplay between magnetic and kinetic pressures drives a stable, periodic Z-pinch behavior accompanied by a recurring “bouncing” effect, which is closely associated with the properties of the current pulse. Specifically, a higher peak current and narrower pulse width result in more frequent pinching events and “bouncing” cycles, leading to enhanced pinching amplitude, plasma density, temperature, and kinetic energy density. However, the introduction of initial density disturbances brings about Magneto-Rayleigh–Taylor (MRT) instability, resulting in the deformation of cylindrical symmetry and earlier termination of Z-pinch processes. Moreover, it is also observed that an increase in disturbance amplitude, as well as the pulse width, exacerbates the evolution of MRT instability, thereby reducing the efficiency and longevity of the Z-pinch process.

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I. INTRODUCTION

Plasma Z-pinch, characterized by the compression of plasma along its axis via the Lorentz force, has emerged as a promising scheme for various scientific and technological applications, including fusion research,^{1–5} high-energy density physics^{6–8} and advanced propulsion systems.^{9,10} It is worth mentioning that Z-pinch processes generating dense and hot plasma can be utilized

to build up powerful and brilliant radiation sources in the extreme ultraviolet and even shorter wavelength range,^{11–16} which can be used for nanoscale imaging and structuring applications in diversified industrial orientations.¹⁷ Accordingly, the discharge-produced plasma (DPP) source^{18–20} was proposed as an indispensable method to generate short-wavelength radiations for inspection and imaging applications.^{21,22} With the fast development of semiconductor industry at present, DPP-based light sources have been extensively

researched and developed worldwide, including the capillary discharge-produced light source,²³ plasma focus light source,^{24,25} hollow cathode-triggered light source,^{26,27} laser-assisted discharge-produced light source,^{28,29} and the electrodeless discharge-produced light source.^{30–33}

Unfortunately, it is inevitable to generate significant thermal loads and electrode corrosion in industrial applications of the discharge-produced light sources. Therefore, frequent cleaning and replacement of the electrodes^{34–37} are required to ensure optimal performance and longevity of the system. Meanwhile, there is always a large amount of debris accompanied by the discharging plasma, which will lead to severe damage to the optical collection system.^{38,39} In contrast, the inductively driven plasma light source proposed by Energetiq is outstanding for its electrodeless design, which reduces the generation of thermal loads on electrodes and associated optical debris effectively.^{30–33} Specifically, the Z-pinch process occurs as a result of the induced current flowing through the plasma source. The induced azimuthal magnetic field results in the magnetic piston compression of the plasma column, facilitating energy conversion from plasma kinetic energy to thermal, plasma ionization, and radiation energies.

However, despite the plasma Z-pinch being an efficient means of energy coupling for power generation, due to insufficient energy conversion efficiency from electric energy to final optical radiation,^{40,41} the inadequate source power remains a crucial challenge for high-volume manufacturing of DPP light sources. In addition, it has found that the quantity and spectra of DPP-based light sources are highly sensitive to Z-pinch dynamics.^{42–44} In addition, as most Z-pinchs are integral parts of pulsed power systems, their dynamic behavior is largely influenced by the performance of the power supply, including parameters like pulse width, pulse energy, and the pulse-to-pulse stability. Moreover, due to the inherent non-uniformity in plasma distribution, instabilities pose a significant challenge in plasma Z-pinchs, which deteriorate the implosion symmetry and the performance of the pinched sources.^{45,46} Therefore, understanding the underlying mechanisms of Z-pinch dynamics and identifying influential factors in the process are imperative for improving both energy conversion efficiency and the performance of the light source.

In this paper, we present magnetohydrodynamic (MHD) simulations to investigate the Z-pinch dynamics of the plasma source operating in the electrodeless inductively driven light source. The work focuses mainly on examining the macroscopic behavior and instabilities exhibited by a current-carrying plasma column during the pinching process, along with a comprehensive analysis of how pulse parameters and density disturbances of the plasma source affect the dynamics. The remainder of this paper is organized as follows: Sec. II outlines the theoretical framework and the two-dimensional (2D) numerical approach used in the study, including the principles of dynamic Z-pinch and the 2D axisymmetric simulation scheme in the r - z plane. Section III presents the simulation results and accompanying discussions, covering topics such as the plasma Z-pinch dynamics, magneto-Rayleigh–Taylor (MRT) instabilities due to initial density perturbations, and the effects of pulse width and amplitude on Z-pinch behavior and MRT evolution. Finally, Sec. IV provides a summary of the key findings and conclusions of the study.

II. THEORETICAL BASIS AND NUMERICAL SCHEME

A. Theory of dynamic Z-pinch

The conventional development scenario of Z-pinch consists of two stages. First, plasma is generated through breakdown, rapid ionization and heating of an initially cold gas, fibers, or wire materials in the first stage. The physical effects governing this stage include avalanche ionization cross sections and other individual characteristics of the ionized medium atoms. Moving on to the next stage, the plasma column is completely ionized and considered to function as an ideal conducting fluid, with an axial current density $\mathbf{J}$ and a resulting azimuthal magnetic induction $\mathbf{B}$. Subsequently, the $\mathbf{J} \times \mathbf{B}$ force, acting on the plasma, forces the column to contract radially. This radial constriction of the plasma column is known as the pinch effect.^{47,48}

Although the equilibrium pinch is always considered when the plasma column radius is constant in time or when it is varying slowly compared to the time required for the plasma to attain a constant temperature. However, static or quasi-static situations do not arise. It is necessary to consider the dynamic behavior of the pinch effect in actual practice. The essential dynamic features of the Z-pinch can be comprehended through the “plasma column” depicted in Fig. 1, i.e., a fully ionized plasma fills the interior region ($0 < r < R_0$) of a hollow dielectric cylinder with radius R_0 and length L , and a current $I(t)$ flows down the plasma column.

Assuming perfect conductivity of the plasma column, all the current will be confined to the surface, resulting in the absence of magnetic flux inside. Therefore, the azimuthal magnetic induction $B_\theta(r)$ induced by the current $I(t)$ can be expressed as⁴⁸

$$B_\theta(r) = \frac{\mu_0 I(t)}{2\pi r}, \quad (1)$$

of which $I(t)$ is the total axial current at the instant t , and r is the corresponding radius of plasma column, with $r = R_0$ at $t = 0$. Then, the magnetic pressure $p_m(r)$ produced by this magnetic induction can be expressed as

$$p_m(r) = \frac{B_\theta^2(r)}{2\mu_0} = \frac{\mu_0 I^2(t)}{8\pi^2 r^2}, \quad (2)$$

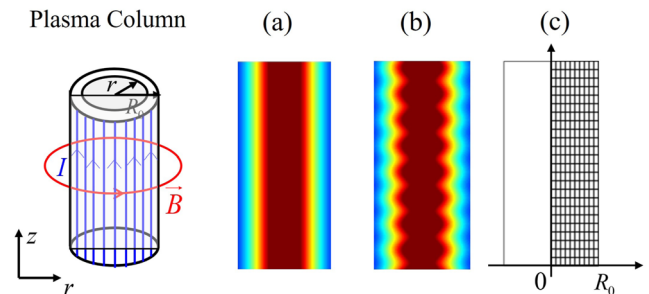


FIG. 1. Schematic diagrams depicting initial density profiles for (a) the undisturbed and (b) the axially disturbed plasma columns in 2D MHD simulations, and (c) the 2D axisymmetrical grid pattern in the r - z plane.

and the radial force acting on per unit length of the current sheath can be written as

$$F(r) = \hat{r}F(r) = -2\pi R p_m(r) \hat{r} = -\frac{\mu_0 I^2(t)}{4\pi R} \hat{r}. \quad (3)$$

To establish the equation of motion, we take the snowplow model into consideration, which assumes that the current sheath carries along with it all of the material it encounters as it moves inward. For the initial mass density ρ_m of plasma column in Fig. 1, the mass per unit length carried by the current sheath moving inwardly can be given by

$$M(r) = \pi(R_0^2 - r^2)\rho_m. \quad (4)$$

According to Newton's second law, there will be

$$\frac{d}{dt} \left[M(r) \frac{dr}{dt} \right] = F(r). \quad (5)$$

Utilizing Eqs. (3)–(5), it can be rewritten as

$$\frac{d}{dt} \left[\pi \rho_m (R_0^2 - r^2) \frac{dR}{dt} \right] = -\frac{\mu_0 I^2(t)}{4\pi R}, \quad (6)$$

which illustrates the temporal dependence of the plasma column radius $r(t)$ on the pinch current $I(t)$.

It is worth noting that this simplified model has disregarded the contribution of kinetic pressure (i.e., thermal pressure) exerted by the plasma column, leading to a gradual reduction of radius $r(t)$ over time. However, it is imperative to consider the significant role played by kinetic pressure in governing the dynamics of plasma Z-pinch. Specifically, as the current sheath moves inward radially and compresses the plasma column, an increase in kinetic pressure must occur. Simultaneously, a dynamic competition arises between magnetic pressure and kinetic pressure that alternates periodically, giving rise to dynamic Z-pinch processes.

B. 2D numerical scheme

In this study, we propose a numerical scheme to investigate the dynamic behavior of plasma Z-pinch due to competition between the kinetic and thermal pressure based on the inductively driven Electrodeless Z-pinchTM light sources. The pulsed current inducing the pinch effect is generated through electromagnetic induction and flows within the discharge-produced plasma column.^{30–33} Then 2D MHD simulations are carried out based on the widely adopted ideal one-fluid model that solves equations of mass and charge continuity, equations of motion and electromagnetic induction, and the one describing ideal gas state, as is listed below:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{v} = 0, \quad (7)$$

$$\frac{\partial \rho q}{\partial t} + \nabla \cdot \mathbf{J} = 0, \quad (8)$$

$$\rho \frac{d\mathbf{v}}{dt} = -\nabla p + \mathbf{J} \times \mathbf{B}, \quad (9)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \quad (10)$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}), \quad (11)$$

$$\frac{d}{dt} (p \rho^{-\gamma}) = 0. \quad (12)$$

During the simulation, the plasma is treated as a compressible ideal fluid without viscosity, and confinement is achieved by an azimuthal magnetic field induced by the pulsed current flowing axially through it.

Similar to cases of gas puffs and high wire number pinches, we employ a two-dimensional (2D) axisymmetric MHD model in cylindrical geometry with a single-fluid plasma approximation in this study. The focus of our study lies in investigating the macroscopic behavior, overall dynamics, and stability of plasma Z-pinch of the discharge-produced plasma sources used for inductively driven light sources. In addition, the simulations are performed in the r - z plane (see Fig. 1) by numerically solving the governing equations Eqs. (7)–(12) using the Finite Volume Method (FVM) with the USim code.

The 2D numerical scheme is illustrated in Fig. 1, where the plasma column is approximated as a density modulated cylinder. Panel (a) represents the initial density profile of the plasma column in the r - z plane, which is distributed in a radial Gaussian profile and has no axial disturbance. Panel (b) depicts the initial density distribution of the plasma column, which possesses both radial Gaussian distribution and axial sinusoidal disturbance. The corresponding density profile can be expressed as $n = n(r) \cdot [1 + a \cdot \sin(2\pi z/\lambda)]$, with $n(r)$, a and λ denoting the initial radial profile, density perturbation amplitude and perturbation wavelength, respectively, while $1/\lambda$ corresponds to the wave number per unit length.

The corresponding numerical grid division is illustrated in Fig. 1(c). Unless otherwise specified, all MHD simulations below are performed within a geometrically sized simulation region of $L_r \times L_z = 400 \mu\text{m} \times 2 \text{ mm}$, corresponding to a grid division of $N_r \times N_z = 256 \times 512$ along the radial and axial directions, respectively. Throughout the simulation, periodic boundary conditions are applied at boundaries along $z = 0$ and $z = L_z$. Meanwhile, the “noInflow” boundary condition is applied at the outer boundary $r = R_0$, implying fluid removal from the simulation upon reaching this rightmost boundary.

In addition, in the case of undisturbed density distribution, the density profile of the plasma column is in the form of $n(r) = n_0 \cdot e^{-r^2/\sigma_r^2}$, with $\sigma_r = 400 \mu\text{m}$. The pulsed current follows a Gaussian profile $I(t) = I_0 \cdot e^{-(t-t_0)^2/\sigma_t^2}$, with I_0 indicating the peak value (i.e., amplitude) of the current pulse. Furthermore, considering the limited sustainability of high energy density in a high-temperature plasma pinch, which is only achievable for brief durations within the sub-microsecond or nanosecond range, our study primarily focuses on investigations pertaining to pulse durations within the nanosecond time frame.

III. SIMULATING RESULTS AND DISCUSSIONS

Now we present two-dimensional (2D) MHD simulations to investigate the Z-pinch dynamics of a current-carrying plasma column and explore the impact of pulse properties and disturbed plasma distributions on the dynamics. The results are presented and

discussed comprehensively to gain insights into the improvement of the performance of DPP-based light sources and the energy conversion efficiency from power supply to optical radiation, which are intricately linked with the dynamic behavior of the plasma Z-pinch in the DPP sources.

A. Z-pinch dynamics of plasma source

First, we conduct 2D MHD simulations to study the Z-pinch dynamics of the plasma column based on the models of Fig. 1. The initial density distribution is given by $n(r) = n_0 \cdot \exp(-r^2/r_0^2)$, where $n_0 = 1 \times 10^{17}/\text{cm}^3$, $r_0 = R_0 = 0.4$ mm, with an initial plasma temperature $T = 1$ eV. In addition, an induced current pulse $I(t) = 5000e^{-(t-50)^2/20^2}$ with the time t normalized to the unit of nanosecond (ns) is employed. The dimensions of the plasma column are $L_r \times L_z = 400 \mu\text{m} \times 2$ mm, corresponding to a cell count of 256×512 . The evolution of the Z-pinch process induced by the pulsed current in the plasma column is shown in Fig. 2, which presents the temporally varied spatial density distribution of the plasma column in the axisymmetrical r - z plane ranging from $t = 0$ to 75 ns. In addition, the corresponding current pulse profile normalized by $I_0 = 5000$ A is shown in panel (a).

It shows that, at the beginning, a slight compression occurred within the plasma column, since the kinetic pressure was large enough to resist the magnetic pressure resulting from the small current pulse. As the current increased over time [see Fig. 2(a)], there was a rapid rise in magnetic pressure $p_B \propto B^2/2\mu_0$, resulting in further compression of the plasma column and formation of a dense plasma shell moving inward radially, as is shown in the panel labeled $t = 25$ ns. Meanwhile, both plasma density and temperature experienced an increase along with the current pulse. Subsequently, the plasma shell was compelled to contract and inflate by the interplay between magnetic pressure and kinetic pressure, leading to

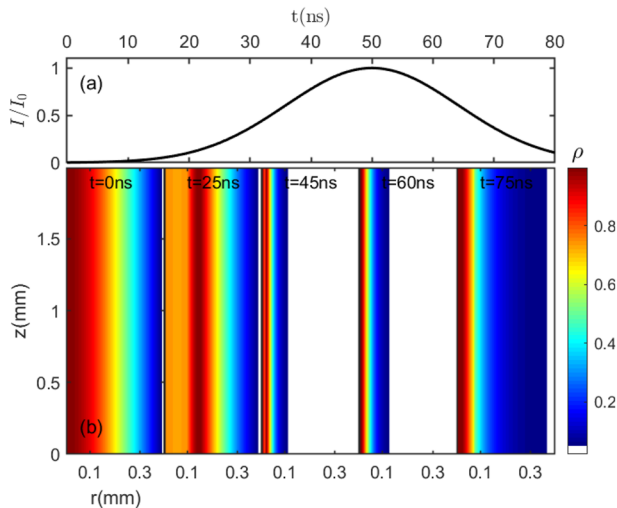


FIG. 2. Temporal evolution of the plasma column density ranging from $t = 0$ to 75 ns, with (a) displaying the temporal profile of the current pulse normalized to I_0 and (b) exhibiting the corresponding evolution of the spatial density distribution of the plasma column.

varying radii of the plasma column, as is shown in panels labeled $t = 45$ ns, 60 ns, and 75 ns in Fig. 2.

The corresponding propagation of the plasma shell is depicted in Fig. 3, which is represented by the variation of the plasma density profile along the $z = 1$ mm axis. It can be observed that the plasma shell moved inward and carried most of the plasma it encountered along with it, resulting in an increasingly denser plasma shell, as is elucidated by the snowplow model.⁴⁸ Afterward, the plasma column started to implode, leading to a decrease in peak density. Meanwhile, a shock motion was created by the current layer (plasma sheath) at the outer radius of the imploding plasma column, which acted as a piston moving inward. That is, there is a high pressure region produced, accompanied by a shock wave propagating ahead of the magnetic piston, which is in good agreement with Refs. 46 and 49.

To get further insight into the dynamics of the Z-pinch process, Fig. 4(a) displays the temporal evolution of plasma density along the axis $z = 1$ mm in a three-dimensional waterfall configuration, while panel (b) displays the corresponding temporal evolution of the peak density of the plasma column. The results show that, since the pinch effect was initiated, the kinetic pressure resulting from thermodynamics counteracted to impede constriction of the plasma column, while the magnetic pressure acted as confinement to impede inflation of the plasma column. Therefore, a quasi-periodic dynamic Z-pinch process was achieved, and the radius of the plasma column changed along with its contraction and dilation due to competition

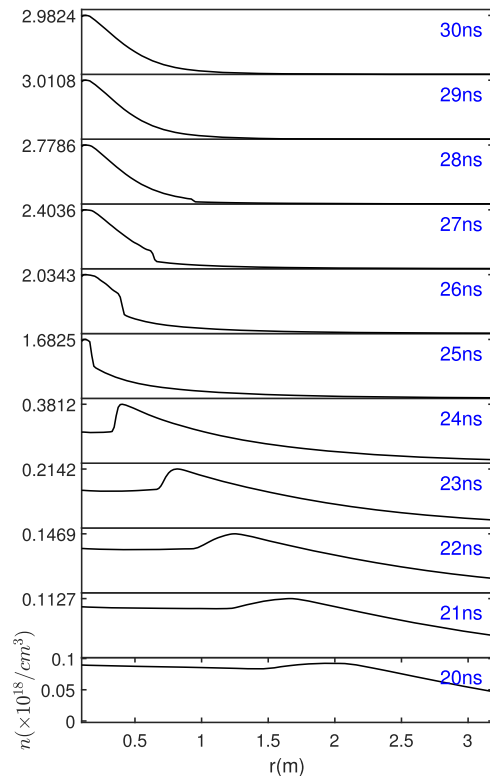


FIG. 3. Motion of the plasma sheath from $t = 20$ to $t = 30$ ns, which is represented by the temporal variation of the plasma density n along the axis $z = 1$ mm in the plasma column.

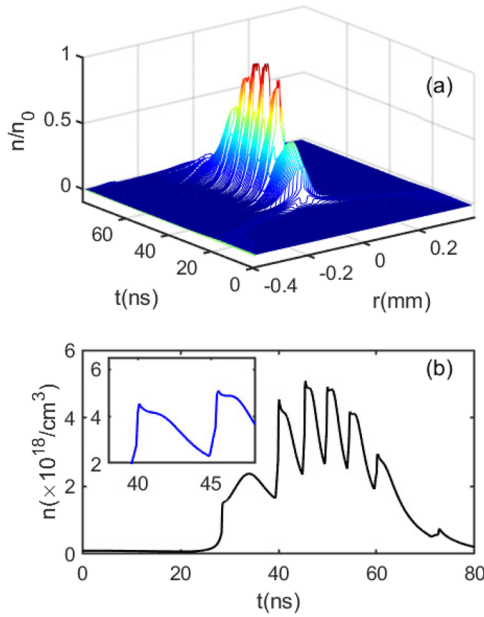


FIG. 4. Variation of the plasma column density during the Z-pinch process, which is indicated by the density profile along the axis $z = 1$ mm, with (a) depicting the density evolution in the t - r plane via a 3D waterfall plot and (b) illustrating the corresponding temporal evolution of the peak density.

between the magnetic pressure and kinetic pressure. In addition, we can find the peak density of the pinched plasma column increases for larger magnetic pressure during contraction and decreases for larger kinetic pressure during thermal expansion. Meanwhile, the pinch period is shortened as the current pulse increases, and vice versa.

In addition, the propagation trajectory of the shock wave generated during the pinching process is shown in Fig. 5, of which panel (a) shows a projection of Fig. 4(a) in the t - r plane, as is experimentally observed in Ref. 50. In addition, the shock trajectory is depicted in a dotted line in panel (a) and illustrated with its interpolation in Fig. 5(b). Specifically, accompanied by the current sheath moving radially inwards, a wave motion was set up to travel inwards first, then get reflected off the axis and moved outward, striking the interface and retarding the inward motion of the current sheath or even reversing it, which is known as “bouncing.”⁴⁸ Meanwhile, we can find the sequence of this bouncing takes place periodically, and the amplitude of each succeeding bounce varies in close association with the temporal profile of the current pulse: the larger the amplitude of the current pulse was, the smaller the amplitude and periodicity of the bouncing became. This attributes mainly to the fact that an increased current pulse induces an intensified magnetic field, then leads to higher magnetic pressure, and then more profound suppression of the thermal expansion caused by kinetic pressure, and vice versa.

In addition, Fig. 6 illustrates the corresponding temporal variations of the normalized thermal pressure (P_T/P_{Tm}), plasma temperature (T/T_m), kinetic energy density (ϵ_k/ϵ_{km}), and magnetic energy density (ϵ_m/ϵ_{mm}) of the plasma column using a dual y -axis scheme.

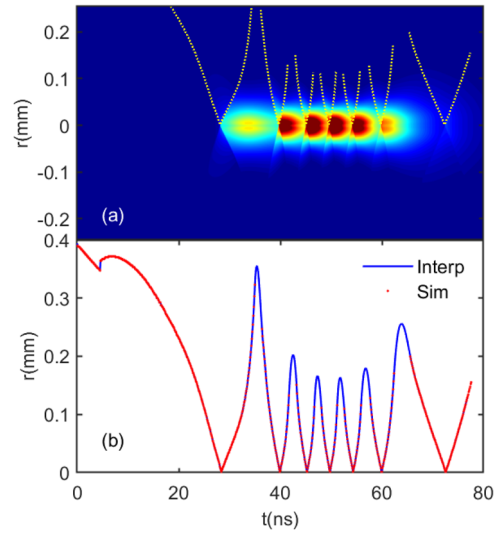


FIG. 5. Propagation of shock wave within the plasma column during the Z-pinch: (a) shows the propagating trajectory of the shock wave in the t - r plane via the 2D density evolution of the plasma column; (b) depicts the interpolation (labeled as *Interp*) of the simulated trajectory (labeled as *Sim*) of the shock wave, which visualizes its “bouncing” process during the Z-pinch.

Among them, P_{Tm} , T_m , ϵ_{km} , and ϵ_{mm} represent the peak values of the plasma’s thermal pressure P_T (Pa), temperature T (K), kinetic energy density ϵ_k (J/m^3), and magnetic energy density ϵ_m (J/m^3), respectively, indicating that each quantity is normalized by its corresponding maximum value. Figure 6(a) demonstrates the variation of plasma temperature with time, which is in consistency with the density profile presented in Fig. 4(b). It shows that an increasing plasma

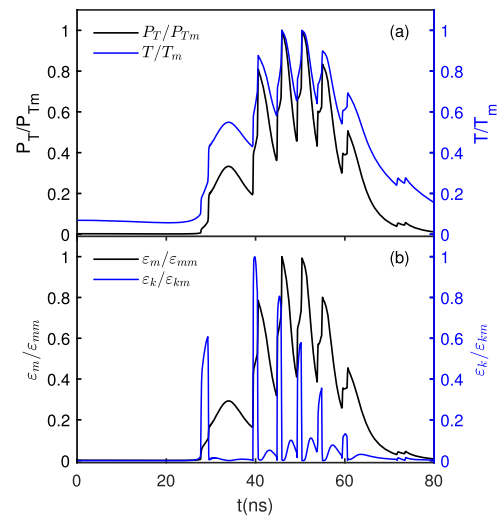


FIG. 6. (a) Temporal evolution of the normalized plasma temperature T/T_m and thermal pressure P_T/P_{Tm} . (b) The corresponding evolution of normalized magnetic energy density ϵ_m/ϵ_{mm} and kinetic energy density ϵ_k/ϵ_{km} during the Z-pinch process.

density results in an increasing plasma temperature, as well as an increasing kinetic pressure (calculated as $P_T = nk_B T$) of the plasma column. Meanwhile, the interplay between magnetic energy density ($p_B = B^2/2\mu_0$) and kinetic (thermal) pressure results in a dynamic Z-pinch, leading to the quasi-periodic variation of magnetic energy density $\varepsilon_m/\varepsilon_{mm}$ depicted in Fig. 6(b). It also reveals that a rapid change in plasma density corresponds to an urgent radial motion of the plasma shell, along with an immediate alternation in kinetic energy density $\varepsilon_k/\varepsilon_{km}$. However, the thermal and magnetic energies of the plasma column are dominant during stagnation periods.

Moreover, it is noteworthy that despite the presence of a radial density gradient, the evolution of the MRT instability in the radially perturbed plasma column has been effectively suppressed. This can be understood as follows. In the case of MHD, where the magnetized two-layered fluid is uniformly distributed within each layer and there is no background flow, the dispersion relation of the system can be expressed as⁵¹

$$\omega^2 = -\frac{kg(\rho_{10} - \rho_{20})}{\rho_{20} + \rho_{10}} + \frac{(\vec{k} \cdot \vec{B}_{20})^2 + (\vec{k} \cdot \vec{B}_{10})^2}{\mu_0(\rho_{20} + \rho_{10})}. \quad (13)$$

Compared to the dispersion relation of Rayleigh–Taylor instability in hydrodynamics (HD), which can be mathematically formulated as⁵¹

$$\omega^2 = -\frac{kg(\rho_{10} - \rho_{20})}{\rho_{20} + \rho_{10}}, \quad (14)$$

we can find the presence of a magnetic field conditions the onset of RT instability in magnetized fluid. Specifically, a certain density difference is required to initiate the MRT instability in MHD, which can be mathematically expressed as

$$\frac{kg(\rho_{10} - \rho_{20})}{\rho_{20} + \rho_{10}} > \frac{(\vec{k} \cdot \vec{B}_{20})^2 + (\vec{k} \cdot \vec{B}_{10})^2}{\mu_0(\rho_{20} + \rho_{10})}. \quad (15)$$

It implies that $\vec{B}$ plays a stabilizing role due to the generated Lorentz force directed radially inward within the plasma column, which tends to counteract the penetration between fluids of different densities.

B. Influence of density disturbance on the Z-pinch dynamics

Although it is possible to achieve a dynamic equilibrium state for plasma confinement with the pinch effect, this state is not stable. Even a slight deviation from the cylindrical geometry of the equilibrium state will lead to amplification of the initial perturbations over time and eventual disintegration of the plasma column. However, non-uniformity in plasma distribution remains an inherent challenge in plasma Z-pinches. In the following, we investigate the influence of density disturbances on the Z-pinch dynamics by subjecting the Gaussian profile of the radial density distribution to an axial perturbation. That is,

$$n = n(r) \cdot [1 + a \cdot \sin(2 \cdot \pi \cdot z/\lambda)], \quad (16)$$

of which $1/\lambda$, a , and n_r indicate the wave number in units of m^{-1} , the initial disturbance amplitude, and the initial radial density

distribution of the plasma column, respectively. In addition, the driving current flowing through the plasma column also follows the Gaussian profile, i.e.,

$$I(t) = I_0 \cdot e^{-(t-t_0)^2/\sigma_t^2}. \quad (17)$$

Figure 7 depicts the temporal evolution of the spatial density distribution of the plasma column, which is initially perturbed by an axial density modulation defined in Eq. (16) with $a = 0.1$, $\lambda = 0.2$ mm. In addition, the driving current presented in panel (a) of Fig. 7 is in the form of Eq. (17), with $I_0 = 5000$ A, $t_0 = 50$ ns, and $\sigma_t = 20$ ns. It can be observed that, as an identical current pulse flows down the plasma column, dynamic contraction and inflation also occur in a similar way to Fig. 2. However, the existence of axial density disturbance triggers the onset of MRT instability, which exhibits significant growth during the dynamic Z-pinch process. Consequently, the MRT instability transitions from a linear stage to a nonlinear phase, resulting in the generation and breaking-through of bubble structures, as well as the bubble-strikes, as is shown in panels labeled “ $t = 50$ ns” and “ $t = 55$ ns.” Furthermore, it can be deduced from Fig. 7 that the MRT instability seeded by the axial density disturbance facilitates the rapid and efficient assembly of a hot and dense plasma along the axis. In other words, the MRT instability in this scenario eventually leads to the formation of localized regions with higher density and temperature, commonly referred to as “hot spots” or “bright spots,” which is in good agreement with the numerical and experimental study of Ref. 52.

Then Fig. 8 illustrates the density evolution of an initially disturbed plasma column during Z-pinch. In contrast to Fig. 2, it shows that the existence of axial density perturbations leads to earlier termination of both the bouncing and Z-pinch processes due to rapid growth of multiple plasma instabilities, including the sausage-type instability⁵³ and MRT instability.^{54,55} According to Ref. 53, the influence of the plasma density on the growth of sausage instability shows that at larger edge plasma densities ($n_{ext} > 10^{-2} n_0$), the sausage

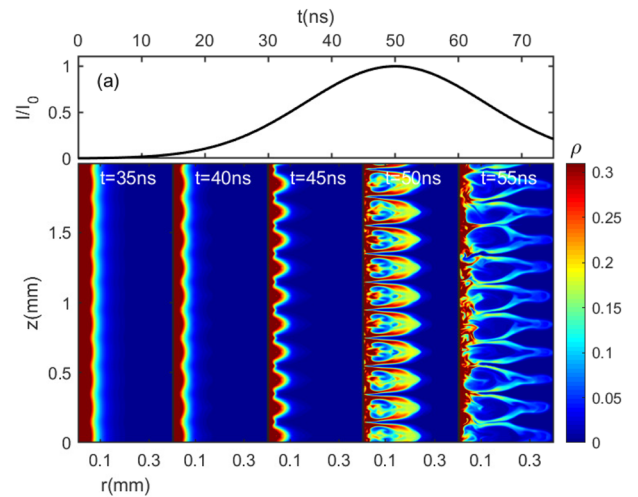


FIG. 7. Temporal evolution of the spatial density distribution of the plasma column, where an initial axial sinusoidal density perturbation with a wavelength of $\lambda = 0.2$ mm is imposed on the radially Gaussian-profile plasma column.

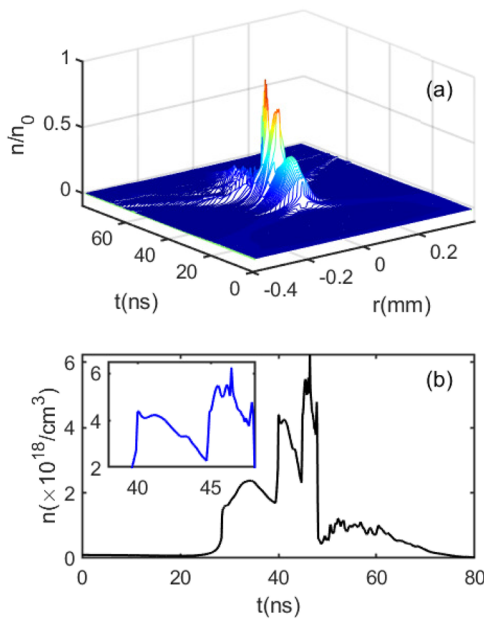


FIG. 8. Density evolution of the initially disturbed plasma column during Z-pinch, which is represented by the density along the axis $z = 1$ mm: (a) illustrates the evolution in the t - r plane via a 3D waterfall plot, and (b) presents the corresponding temporal profile of the peak density.

instability tends to be suppressed because a substantial part of the pinch current begins to flow through the rarefied pinch plasma. In addition, the growth of sausage instability in the case of a current distributed over the pinch evolves slowly and is generally less pronounced in contrast to skin current. Therefore, the instability seeded by the axial density perturbation in Fig. 8 is focused mainly on the MRT instability, which shows significant growth during the period of declining current pulse. This can be dated back to the fact that a decline in the current pulse results in a decrease in magnetic pressure and the corresponding radial confinement, thereby weakening the suppression of MRT instability and enhancing the disruption of the Z-pinch process, especially when the amplitude of perturbation transitions to its nonlinear phase.

In addition, in contrast to Fig. 4(b), we can find the presence of noisy oscillations in the n - t line for the initially disturbed case in Fig. 8(b). The noise can be regarded as high-frequency terms, which is caused by the presence of MRT instability with higher wave numbers during the Z-pinch process, as is depicted in Fig. 9. Here, the evolution of MRT instability is characterized by the temporal variation of the perturbation mode calculated by Fast Fourier Transform (FFT). In addition, each panel showcases the normalized FFT spectra amplitude of the perturbation mode during plasma Z-pinch, spanning from $t = 0$ ns to 80 ns. As we can see, a Fourier decomposition of the density perturbation contains not only the imposed mode of wave number $k = 5000/\text{m}$ but also a number of adjacent modes with smaller amplitudes. The mode evolves to encompass the fundamental and higher-order harmonics, like the second, third, fourth, and so on, declaring an increasingly profound transition of MRT instability from the linear stage to the nonlinear one. Furthermore, as

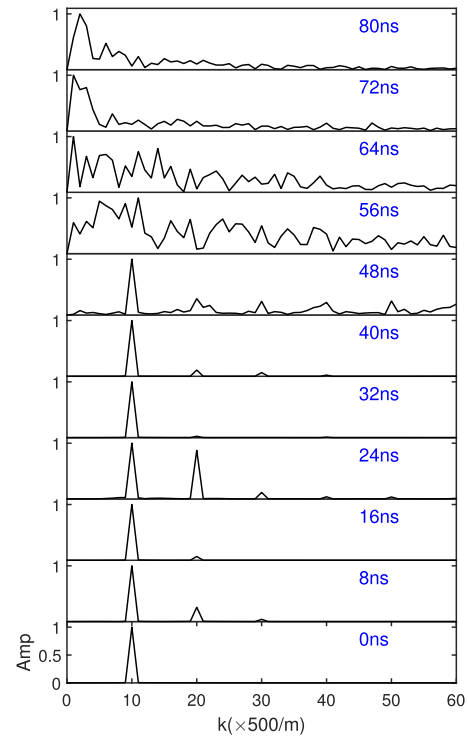


FIG. 9. Temporal evolution of the perturbation modes characterized by FFT spectra of the axial density distribution. Here, *Amp* indicates the normalized amplitude of the spectra by its peak, and k indicates the wave number, whose value is $k = 5000/\text{m}$ for the initial perturbation.

bubble formation and breakthrough occur, an increasing number of wave modes contribute to the formation of continuous FFT spectra, as shown in panels labeled “48–80 ns.”

The influence of the initial disturbance amplitude a on the dynamics of the Z-pinch is shown in Fig. 10. The pulsed current is profiled as $I(t) = I_0 \cdot e^{-(t-t_0)^2/\sigma_t^2}$, where $I_0 = 5000$ A, $t_0 = 50$ ns, and $\sigma_t = 20$ ns are selected for all of $a = 0, 0.1, 0.3$, and 0.5 . In addition, $a = 0$ denotes the absence of any initial axial density disturbance. The upper panels in Fig. 10 present variations in the density profiles of the pinched plasma column in the r - t plane, while the lower panels present the corresponding temporal evolutions of peak densities.

The result suggests that an increase in initial perturbation amplitude results in a reduced efficiency of the pinch effect. Although a certain perturbation could cause a localized increase in plasma density and temperature during the pinching process (see panels labeled “ $a = 0.1$ ”), the maximum peak value of the pinched plasma column decreases with increasing perturbation amplitude a , as well as the temperature of the pinched plasma. This can be understood as follows. Generally, the amplitude of MRT instability grows linearly until it reaches a threshold where it transitions into the nonlinear stage. However, increasing perturbation amplitude accelerates the transition of MRT instability into the nonlinear stage, which enhances the nonlinear coupling between different perturbation modes and bubble evolution. Consequently, reduced

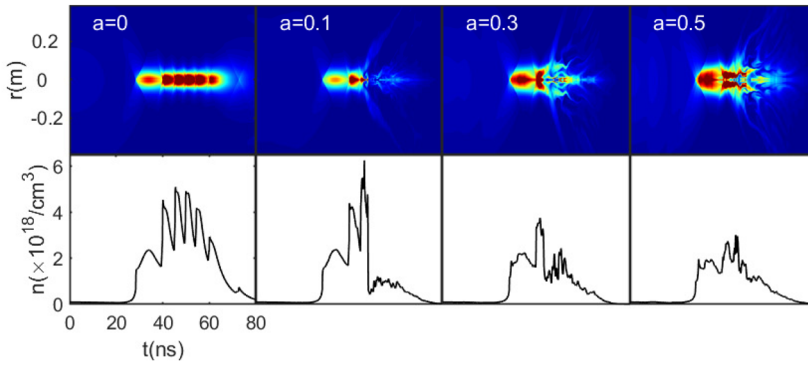


FIG. 10. Influence of the initial perturbation magnitude a on the dynamics of the plasma Z-pinch, which is illustrated by the density evolution in the t - r plane and the corresponding temporal variation of the peak density. Here, $a = 0, 0.1, 0.3, 0.5$ are labeled, respectively, with $a = 0$ denoting the case of unperturbed density distribution.

performance and weakened stability of the Z-pinch process are observed in Fig. 10.

C. Influence of pulse width and amplitude on the Z-pinch dynamics

In this section, we investigate the dynamics of the Z-pinch for different driving current pulses based on models (a) and (b) in Fig. 1. The initial density distribution and driving current pulse follow the forms of Eqs. (16) and (17), respectively.

Figure 11 illustrates the influence of pulse amplitude on the evolution of Z-pinch dynamics and MRT instability. Here an initial perturbation with a wave number $k = 5000/\text{m}$ (corresponding to $\lambda = 0.2 \text{ mm}$) and an amplitude $a = 0.1$ is employed. In addition, the pulse amplitudes are selected to be $I_0 = 3000, 5000$, and 9000 A for Eq. (17) with $t_0 = 50 \text{ ns}$ and $\sigma_t = 20 \text{ ns}$, as is depicted in Fig. 11(b). The corresponding temporal variations of the plasma column density are shown in Fig. 11(a) by the temporal evolution of peak density

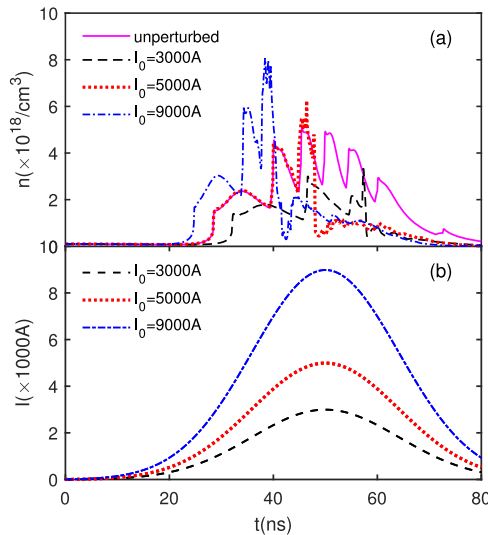


FIG. 11. Influence of the current pulse amplitude on Z-pinch dynamics: (a) presents the variation of peak densities and (b) presents the corresponding pulse profiles for different amplitudes. The label “unperturbed” refers to the case of unperturbed density distribution with $I_0 = 5000 \text{ A}$.

along the axis $z = 1 \text{ mm}$. The results show us an intensified pinching effect and increased plasma density due to the enhancement of magnetic compression of the plasma column with increasing current pulse amplitude. Meanwhile, the pinching (bouncing) period is shortened for increased current pulse in Fig. 11(a), which implies an augmented implosion and bouncing velocity. In addition, it is worth noting that the MRT instability induced by density disturbance has been further amplified by the increasing current amplitude, resulting in earlier onset and disruption of the Z-pinch process.

Then the influence of current pulse width on Z-pinch dynamics and the evolution of MRT instability is illustrated in Fig. 12. Figure 12(a) shows the results for a current pulse profiled as $I = I_0 e^{-(t-50)^2/20^2}$, while Fig. 12(b) presents those for $I = I_0 e^{-(t-100)^2/50^2}$, where I_0 is set to 9000 A in both cases. The dashed lines labeled “ $I(t)$ ” indicate the temporal profiles of normalized current pulses, which are scaled by the right y -axes. Meanwhile, the temporal evolution profiles normalized by $n_m = 1 \times 10^{18}/\text{cm}^3$ are scaled by the

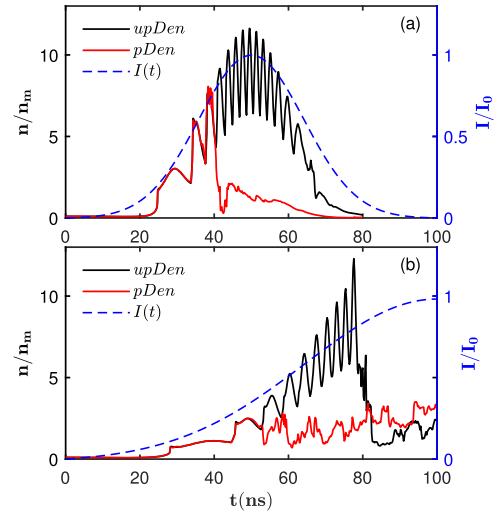


FIG. 12. Influence of the pulse width on Z-pinch dynamics illustrated in dual y -axis scenarios, of which panels (a) and (b) display the temporal profiles of the current pulses in blue dotted lines [labeled “ $I(t)$ ”], and the temporal evolution of plasma densities with (labeled “pDen”) and without (labeled “upDen”) initial perturbations, respectively.

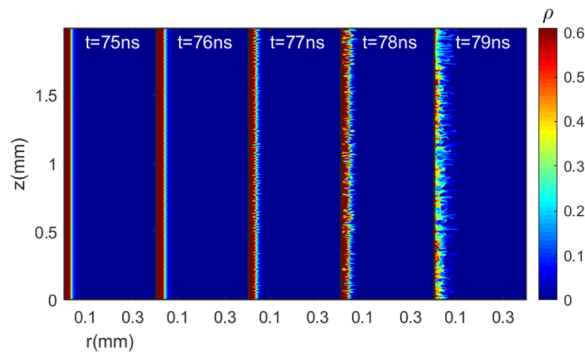


FIG. 13. Influence of the pulse width on the development of Z-pinch instability for the plasma column without initial axial perturbation.

left y -axes, with “pDen” and “upDen” indicating cases with and without initial density perturbation, respectively. The results show that an increase in pulse width reduces the efficiency of plasma Z-pinch and enlarges its period of bouncing. In addition, Figs. 12(a) and 12(b) indicate that the stability of dynamic Z-pinch and its longevity are also reduced by an increased pulse width, regardless of whether the plasma source is initially perturbed or not.

The reasons can be explained from the following perspectives. First, the plasma column acts as an impedance from an electrical view; the induced electromotive force $\mathcal{E}$ is proportional to the value of $dI(t)/dt$. Obviously, an increase in pulse width will result in a decrease of $dI(t)/dt$, leading to a reduction of $\mathcal{E}$, as well as the magnetic confinement. Second, a higher magnitude of the current pulse leads to an increase in both $\mathbf{v}$ and $\mathbf{B}$, thereby enhancing the magnetic confinement and radial compression of the plasma column through the augmented Lorentz force ($\mathbf{F} \propto \mathbf{v} \times \mathbf{B}$). Therefore, the widened pulse width enhances the development of MRT instability seeded by initial axial perturbation. Furthermore, despite the unperturbed initial density distribution n_r , the extended pulse width contributes to generating a radial density gradient at the outer boundary of the plasma column, as is shown in Fig. 13. Consequently, according to Eq. (15), a sufficient radial density difference forms at the density interface, leading to MRT instability and the rapid collapse of the dynamic Z-pinch process.

IV. CONCLUSIONS

The plasma Z-pinch is a highly efficient mechanism for generating high-energy-density plasma sources in inductively driven light systems. Gaining deeper insights into the dynamics of the Z-pinch process is of great significance for enhancing the performance of DPP-based light sources. In this study, 2D MHD simulations based on the axisymmetrical scheme have been conducted to investigate the Z-pinch dynamics of the current-carrying plasma column. The temporal and spatial evolution of the plasma column during the pinching process was studied in detail, along with the temporal evolution of the energy density, magnetic pressure, and kinetic pressure. In addition, the influences of current pulse characteristics and density disturbances on the stability and overall performance of the Z-pinch process were analyzed.

Our simulations of the dynamic Z-pinch process have presented the formation of a plasma sheath (current sheath) in the plasma column and yielded good agreement with the dynamic Z-pinch theory, as is shown in Figs. 2 and 3. Meanwhile, the shock motion has been demonstrated to be created by the current layer (plasma sheath) at the outer radius of the imploding plasma column, which acted as a piston moving inward. The periodic pinching and bouncing were found to take place in response to the competition between magnetic pressure and kinetic pressure for varying current pulses in Figs. 4–6. However, the introduction of axial sinusoidal density distribution led to the rapid development of MRT instability and prematurely terminated the efficient Z-pinch process in Figs. 7–9. In addition, a reducing pinching efficiency was observed for increasing perturbation amplitude in Fig. 10.

Moreover, the influence of the current pulse on Z-pinch dynamics has been investigated and illustrated in Figs. 11–13. The results reveal that an increase in current pulse amplitude leads to a shorter period of the pinching and bouncing processes, accompanied by higher implosion velocity, pinching amplitude, plasma column density, and temperature, which is highly desirable for light sources based on the Z-pinch process of DPP sources. However, the intensified pinching process and higher implosion velocity contribute to the rapid development of MRT instability and its transition into the nonlinear stage. Meanwhile, extending the current pulse width reduces the efficiency of the Z-pinch process and enhances the instabilities during pinching. Consequently, the axial symmetry within the plasma column is disrupted, and the Z-pinch process collapses, which is detrimental to its application in DPP-based light sources. Therefore, in order to maintain a stable Z-pinch process and achieve a reliable DPP-based light source, it is imperative to ensure the presence of uniform and undisturbed plasma sources, as well as narrow current pulses with high current values.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yanxia Xu: Investigation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Make Ying:** Project administration (lead); Supervision (lead); Visualization (supporting); Writing – original draft (supporting); Writing – review & editing (supporting). **Xuanyu Yuanhua:** Data curation (equal); Software (equal). **Jianhui Zhou:** Conceptualization (equal); Data curation (equal); Formal analysis (equal). **Xiaoqiang Zhang:** Conceptualization (equal); Formal analysis (equal). **Guanxin Huang:** Conceptualization (equal). **Benkun Lv:** Software

(equal). **Haonan Zhang:** Data curation (equal). **Jiangwei Li:** Project administration (supporting); Supervision (supporting). **Zongqiang Yu:** Project administration (supporting); Supervision (supporting). **Yang Yang:** Formal analysis (equal); Software (equal). **Yue Cheng:** Writing – review & editing (supporting). **Xijun Li:** Writing – review & editing (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

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